

Developer Genome Project Analysis

Project overview:

Developer Genome is a full-stack developer analytics platform that turns GitHub activity into an evolving technical and behavioral profile. The system collects repository signals, computes DNA scores, tracks historical trends, and presents the results in an interactive dashboard.

Core capabilities:

- GitHub-inspired data model and activity aggregation
- Python analytics engine for technical and behavioral scoring
- Historical trend snapshots for milestone tracking
- Responsive dashboard with radar charts, score evolution, repo health, and technology mix
- GitHub Actions automation for recurring refreshes
- Public-ready frontend architecture for deployment to Vercel or Netlify

Current implementation status:

- Backend analytics engine is built and generates a sample snapshot
- Dashboard frontend is implemented in React + Vite with polished dark/light styling
- Sample analytics data is committed in `backend/data/analytics_snapshot.json`
- Daily automation workflow is prepared under `.github/workflows`

Verified results:

- Python analytics snapshot generation succeeded
- Frontend production build succeeded with Vite
- Overall score in the generated sample dataset: 61.7
- Technical DNA highlights: Backend 58.0, Data 54.0, Frontend 42.7
- Behavioral DNA highlights: Consistency 84.0, Exploration 85.0

Recommendation:

Integrate the real GitHub API and add per-user authentication, persistent storage, and AI insight generation to scale this into a production-grade public product.